HONGYUAN NIU

Chongqing, CHINA, 400045

draconhy@gmail.com | https://aaronivy0731.github.io | linkedin.com/in/hongyuan-niu

EDUCATION

Chongqing University, *Master of Science in Urban Planning* Sep 2023 – Jun 2026

- **GPA:** 3.7/4.0
- **Coursework:** Spatial Planning Techniques, Urban Governance, Regional Development, Planning Theory
- Admitted as top-ranked applicant in the academic master's track.

Southwest Jiaotong University, *Bachelor of Engineering in Urban Planning* Sep 2018 – Jun 2023

- **GPA:** 3.5/4.0
- **Top scores in core courses**: Urban Sociology, Planning History, Infrastructure Planning, and Urban Economics.
- Graduated as Excellent Graduate and won multiple awards for academic scholarship and leadership.

ESTP, Paris, *exchange* Sep 2024 – Jan 2025

RESEARCH INTERESTS

Urban Public Service System, Spatial Accessibility and Equity, Urban Network Analysis, GIS and Spatial Analysis
Urban Resilience, Healthcare Geography, Nighttime Urban Studies

RESEARCH EXPERIENCE

Research on Nighttime Healthcare Service Network of Chongqing Mar 2025 – Jul 2026
M.S. Thesis, Chongqing University

- Developed time-specific healthcare supply–demand and inter-facility coordination networks for 318 healthcare facilities and 161 subdistricts across 4 nighttime periods, integrating facility opening schedules, population demand, and road traffic conditions
- Identified a shift in dominant network constraints from traffic impedance to late-night supply contraction
- Evaluated targeted spatial and temporal interventions, demonstrating how selective site additions and extended operating hours can improve service coverage and network redundancy

Spatiotemporal Pattern of Cascading Disasters in Mountainous Cities Dec 2023 – Sep 2024
Chongqing Natural Science Foundation - Spatial intervention of cascading failures in lifeline networks

- Examined case evidence from Southwest China to investigate how landslides, debris flows, and floods can trigger cascading disruptions in urban lifeline systems and the built environment
- Summarized the response strategies of disaster prevention and mitigation in mountainous urban planning

Urban Design and Planning Guidelines for Wulong District (Design Project) Jan 2024 – Aug 2024
Chongqing University Research Institute of Architecture and Planning Design

- Conducted field surveys, GIS-based suitability analysis, and stakeholder consultation for district-level urban design and planning guidelines, incorporating feedback from over 10 stakeholder meetings
- Proposed optimization strategies and visualized urban design outcomes in low-altitude economy adaptive planning design and adopted by the local government as a statutory plan

Spatial Productivity in Chengdu's Nighttime Economy Jun 2020 – May 2021
China National Student Research Training Program

- Constructed a 20-indicator framework across four dimensions to examine the spatial determinants of night-market vitality and distribution in Chengdu
- Applied space syntax and regression analysis to identify spatial relationships and develop planning recommendations; the project received the Excellence Prize at the 2022 WUPENicity

PUBLICATIONS & CONFERENCES

Journal Manuscripts:

- **Niu, H.**, & Huang, Y. (2026). Keeping Pace with Urban Growth? Long-Term Spatial Alignment of Tertiary Grade-A Healthcare Resources in Four Southwestern Chinese Cities. ***Land***, 15(10), 1984. https://doi.org/10.3390/land15101984
- **Niu, H.**, & Huang, Y. Spatiotemporal Evolution and Planning Response of Urban Nighttime Healthcare Service Networks: A Case Study of the Central Urban Area of Chongqing. Under review at ***Sustainable Cities and Society***

Selected Conference Publications & Presentations:

2026 AAG Annual Meeting

- **Niu, H.** Spatial Patterns of Nighttime Healthcare Accessibility in a Mountainous City: A Network-Based Analysis of Chongqing.

2025 ACSP Annual Conference

- **Niu, H.**, & Zhang, P. Regenerating Aging Communities via Intergenerational Symbiosis: The Case of Yulin, Chengdu.

2025 ISOCARP World Planning Congress

- **Niu, H.**, & Zhang, Z. Governing Small Town Shrinkage under Depopulation Crises: Global Policy Patterns and Local Pathways for Sustainability in China.

2025 SPSD International Conference

- **Niu, H.**, & Zhang, Z. Spatial Polarization and Governance Strategies for County-Level Digital Villages: An Empirical Study from Four Provinces in Northern China.

2025 The Annual National Planning of China

- **Niu, H.** Incentive-Compatible Strategies for Revitalizing Hollow Villages: Learning from Italy's One-Euro Housing Policy. DOI:10.26914/c.cnkihy.2025.046186
- **Niu, H.** Translating Traditional Planning Principles in Modern Context: A Comparative Study of Beijing's Central Axis and Paris. DOI:10.26914/c.cnkihy.2025.045654

2024 The Annual National Planning of China

- Zhang, P., Zhang, Z., **Niu, H.**, Xiang, M., Pan, Y. Quantitative Evaluation of Urban Night Market Vitality Based on Spatial Data: A Case Study of Central Chengdu. DOI:10.26914/c.cnkihy.2024.035295

SKILLS

Languages: Mandarin Chinese (mother language), English (IELTS : 7), French (A1)

Research Methods: GIS and spatial analysis, complex network analysis, accessibility analysis, spatial modeling and statistics, data visualization

Software: ArcGIS, AutoCAD, SketchUp, Adobe Photoshop, Gephi, Codex, LaTeX